

Orbital History Writes the Cosmic Clock on Dwarf Galaxies: A Pre-Registered, Modified-Gravity-Impossible Test of de Sitter–Unruh Modified Inertia

Carl P. Zimmerman Briar Creek Tech 2026-06-27 — v2 (the memory kernel $\theta(y)$ sharpened from a free function to a bath-derived $\theta(0)=\sqrt{2}$ + Lorentzian shape; the prediction, sign, $y=1$ zero-crossing, and pilot are unchanged)

Abstract

I pre-register a sharp, falsifiable prediction of the de Sitter–Unruh modified-inertia framework — the proposal that a body’s inertia is its **nonlocal-in-time response to the cosmic-horizon Unruh bath**, a bath that carries exactly one clock, the de Sitter rate H_Λ , fixing $a_0 = cH_\Lambda/Z = 9.36 \times 10^{-11} \text{ m/s}^2$. Because inertia in this picture is a functional of the recent acceleration *history*, not of the momentary acceleration, the bath’s clock must imprint on the internal dynamics of any object whose external acceleration *changes* on a timescale comparable to its own internal orbital frequency. The cleanest accessible site is a diffuse Local-Group dwarf galaxy on a radial-plunge orbit: at fixed pericenter distance and fixed mass, a high-eccentricity plunger should run **hotter** (larger internal velocity dispersion σ) than a circular-orbit dwarf of the same mass and same closest approach. The **sign is a theorem** (the memory kernel θ is decreasing, Milgrom-forced, so a plunge sheds adiabatic external loading and drops the dwarf deeper into the deep-MOND regime); the **magnitude is now bath-constrained, not free** — the de Sitter–Unruh correlator *selects* $\theta(0)=\sqrt{2}$ and *forces* a Lorentzian kernel, giving **+12–14%** across the plunging band (a +12–30% envelope spanning the one surviving memory-order residual), with the exact zero-crossing at $y=1$ forced and only the kernel’s corner *location* left as a quasi-static modeling choice. The discriminating physical axis is **$y = \omega_{\text{ext}}/\omega_{\text{internal}}$** , gated by diffuseness, not raw eccentricity. The two named carriers that actually reach the non-adiabatic band are **Crater II** ($y = 3.28$) and **Antlia II** ($y = 2.55$), against adiabatic controls Fornax and Sculptor ($y < 0.3$). Crucially, this prediction is **modified-gravity-impossible**: in any metric or field MOND (AQUAL/QUMOND/AeST) and in Λ CDM, the external-field effect is instantaneous — internal dynamics “depend only on the momentary value of a_{ex} ” (Milgrom 2022) — so those theories predict *exactly zero* σ -vs-history correlation at fixed pericenter, for any a_0 . Only modified inertia gives a nonzero, history-dependent σ ; this joins the Cassini bound as a clean modified-inertia-vs-modified-gravity discriminator. **I report the honest current status straight.** A pre-registered pilot on 24 real Milky-Way dwarfs (Pace, Erkal & Li 2022, Gaia EDR3 orbits) gives a partial Spearman $\rho(\sigma, \text{eccentricity} \mid r_{\text{peri}}, \text{mass}, r_{\text{half}}) = -0.196$, **$p = 0.395$** — a **null**, slightly the wrong sign, robust to a tidal-heating control. This is **not** a falsification and **not** a hint: it is a null with almost no statistical power, because only two of 24 dwarfs reach the carrier band, eccentricity is a noisy surrogate for the real y -axis, and the per-object σ errors are comparable to the predicted signal. Existing data cannot yet test this prediction. The decisive test is Gaia DR4 (December 2026) plus diffuse-carrier spectroscopy and resolved profiles, run as a carrier-vs-control **y-contrast** with explicit

tidal modeling. This is a pre-registered, near-term, MG-impossible prediction — **not a TOE, not a detection.**

1. The framework as subject: inertia as a horizon-bath response

This paper treats one specific physical proposal as its subject and reasons forward from that proposal's own premises. The premise is not a variant of Milgrom's MOND and should not be read through MOND's lens.

The premise is this. An accelerated observer in de Sitter space sees a thermal bath — the Gibbons–Hawking/Unruh radiation of the cosmic horizon (Deser & Levin 1997). The framework takes the next step and asserts that **inertia itself is the body's reaction to that bath**: to resist acceleration is to push against the horizon's radiation. The bath has exactly one intrinsic rate, the de Sitter correlation frequency

$$H_\Lambda = cH_\Lambda/c = 1.81 \times 10^{-18} \text{ s}^{-1}, \quad cH_\Lambda = 5.42 \times 10^{-10} \text{ m/s}^2,$$

set by the dark-energy density alone (the pure- Λ , de Sitter footing). The acceleration scale at which the bath response becomes order-unity is then

$$a_0 = cH_\Lambda/Z = 9.36 \times 10^{-11} \text{ m/s}^2,$$

with Z the framework's geometric normalization. This is a *horizon-derived* acceleration, not a fitted parameter. Above a_0 a body is in the ordinary inertial regime; near and below a_0 the bath response weakens the effective inertia, and a fixed gravitational force produces a larger acceleration — the phenomenology usually labelled MOND, here arising from *modified inertia* with its own interpolation,

$$g_{\text{obs}} = \sqrt{(g_{\text{bar}}^2 + g_{\text{bar}} \cdot a_0)}, \quad \mu_{\text{fw}}(x) = (\sqrt{(1 + 4x^2)} - 1)/2x,$$

(the framework's own excess-heat interpolation; I use this throughout and never McGaugh's v). On steady, hyperbolic-acceleration data (galaxy rotation curves, the radial-acceleration relation) this collapses algebraically to the same curve generic MOND produces, so the RAR cannot, by itself, distinguish the framework — a point I state plainly rather than spin. The framework's *distinctive* content lives where the acceleration history is non-trivial, which is exactly where this paper looks.

For the reader: the key word is **nonlocal-in-time**. Milgrom (1994) proved that any consistent modified-inertia theory must be time-nonlocal — the inertial reaction cannot be a function of the instantaneous acceleration alone; it must be a functional of the worldline carrying a memory kernel. The bath supplies the only clock that kernel can use: $1/H_\Lambda$. A body's inertia therefore reads the *history-averaged* state of its motion. That is the engine of everything below.

2. The prediction: a plunge runs hotter

Consider a diffuse dwarf galaxy orbiting the Milky Way. Two frequencies compete:

- ω_{internal} — the dwarf’s own internal orbital frequency (its stars circulating in its potential), and
- ω_{external} — the rate at which the *external* (host) acceleration the dwarf feels is changing as it moves along its orbit.

Define the dimensionless ratio

$$y = \omega_{\text{external}} / \omega_{\text{internal}}$$

(Milgrom 2022, Eqs. 28/33/34). The memory kernel enters as $\theta(y)$:

- For a **circular** orbit, $|a_{\text{ext}}|$ is constant. The kernel sees a stationary history. The response is purely adiabatic — the constant-external-field limit $\theta(0)$, which is the static external-field-effect “thermometer” reading already used for Crater II in earlier work.
- For a **radial plunge**, $|a_{\text{ext}}|$ sweeps up as the dwarf falls in. Near pericenter ω_{ext} climbs toward ω_{internal} , so $y \rightarrow O(1)$: the response is genuinely **non-adiabatic**, and $\theta(y)$ is a real, varying function rather than a constant.

Now the sign. The kernel θ is **decreasing** — $\theta(0)$ is a few, $\theta(1) = 1$ — a feature forced by Milgrom’s construction, not chosen. A circular dwarf carries the full adiabatic external loading $\theta(0) \cdot g_{\text{ext}}$, which partially pulls it *out* of the deep-MOND regime (this is the ordinary external-field effect that makes satellites a little more Newtonian). A plunging dwarf, momentarily non-adiabatic at pericenter, **sheds** part of that adiabatic loading: $\theta(y < \text{the adiabatic value})$ means less effective external field, so the dwarf drops *deeper* into deep-MOND, its effective inertia falls further, and its stars move faster. **A plunge runs hotter.**

Since σ scales as roughly the quarter-power of the effective regime depth, the boost depends on $\theta(0)$. **In this version the de Sitter–Unruh bath itself partially derives the kernel** (previously a free function). Two framework-internal results pin its core: (i) the bath’s *excess-heat* coupling $\Delta T = T(a) - T(0)$ is degree-1 (an amplitude in acceleration — the framework already discards the energy combination $T^2 - T_{\Lambda}^2$ because it gives the wrong deep-MOND law), and the additive-acceleration construction puts the external field into the first moment at unit weight on a single de Sitter floor, which together **select $\theta(0) = \sqrt{2}$** (not the energy-branch value 2); (ii) the de Sitter Wightman function ($\propto 1/\sinh^2$) has exponential memory, whose transform **forces a Lorentzian kernel** $\theta(y) = \sqrt{2} / [1 + (\sqrt{2} - 1) y^2]$ with a y^{-2} tail. The one piece the bath does *not* fix is the **corner location** $y = 1$ (the bath’s own scale H_{Λ} sits orders below every bound-orbit frequency); placing the corner at the internal orbital frequency ω_{internal} is a quasi-static modeling choice, licensed by Milgrom (1994, Eqs. 55–57), not derived.

With this derived kernel, the predicted member-dispersion boost is

$\sigma_{\text{radial}} / \sigma_{\text{circular}} \approx +12 \text{ to } +14\%$ across the realistic plunging band $y \approx 0.35\text{--}0.50$ (preferred, single-clock $\theta(0)=\sqrt{2}$),

monotone-decreasing in y , with an **exact zero-crossing at $y = 1$** and suppression beyond (-8% at $y=1.3$, -23% at $y=2$). The **existence, sign, and the exact $y=1$ zero-crossing are kernel-independent theorems** (they follow only from θ decreasing through $\theta(1)=1$). The magnitude is now **bath-constrained, not free**: $+12\text{--}14\%$ is the preferred value, with a **$+12\text{--}30\%$ envelope** whose upper end corresponds to the lone surviving residual — a two-pole ($\theta(0)=2$, y^{-4} -suppressed) memory reading that the single de Sitter clock disfavours. This replaces the earlier free-function “factor- ~ 2 ” estimate; the open part of the kernel is now squeezed to the single corner-location postulate.

Why diffuseness gates it. y is large only when ω_{internal} is *small* — i.e. for diffuse, low-internal-frequency dwarfs. A dense dwarf has high ω_{internal} , so even a steep plunge keeps $y \ll 1$ and the response stays adiabatic: dense dwarfs are *immune* and serve as a built-in internal control. Milgrom himself flagged the relevant population: “in some dwarf satellites of the Milky Way and Andromeda we estimate $\omega_{\text{ex}} \sim \omega_{\text{in}}$.” Those diffuse satellites are precisely the carriers. The axis of the test is therefore **diffuseness-gated y** , not raw eccentricity — a distinction that turns out to be decisive for interpreting the pilot (Sec. 5).

3. The modified-gravity-impossibility theorem

What makes this prediction a *discriminator* rather than just another MOND signature is that no modified-gravity theory can produce it, even in principle.

In every metric or field formulation of MOND — AQUAL, QUMOND, and the relativistic completion AeST (the framework’s only covariant home) — the external-field effect is **instantaneous**. Milgrom (2022) states it verbatim: a subsystem’s internal dynamics “depend only on the momentary value of a_{ex} .” The internal dynamics are governed by the value of the external field *right now*, with no memory of how the system arrived there. Therefore, comparing two dwarfs of identical mass at identical pericenter distance — one on a circular orbit, one on a radial plunge — a modified-gravity theory predicts **exactly the same internal σ** , hence

$$\rho(\sigma, \text{orbital history} \mid r_{\text{peri}}, \text{mass}) = 0, \text{ for any } a_0, \text{ in any metric/field MOND.}$$

$\Lambda\text{CDM}/\text{CDM}$ gives the same null for the same reason: a circular and a radial subhalo at matched current radius have the same internal dynamics modulo tidal history (which is the confound, addressed in Sec. 6, not the signal). **Only modified inertia — inertia as the nonlocal bath-response functional — yields a nonzero, history-dependent σ .** The logical structure is identical to the Cassini bound, which excludes the framework’s modified-gravity cousins because *they* would imprint a trajectory/history dependence on solar-system orbits that the data forbid; here the *same* nonlocality appears as a *positive* prediction in a fresh, accessible observable. A zero result is consistent with all of modified gravity and CDM; a nonzero positive σ -vs-history correlation at fixed pericenter would be impossible for any of them and required by the framework.

4. The carriers, the controls, and the diffuseness gate

Pre-specifying carriers on physics (diffuseness), not on σ , the per-dwarf y computation places only two objects firmly in the non-adiabatic carrier band:

dwarf	role	a_{ext}/a_0 @ peri	$y = \omega_{\text{ext}}/\omega_{\text{in}}$	regime
Crater II	carrier	0.46	3.28	NON-ADIABATIC
Antlia II	carrier	0.35	2.55	NON-ADIABATIC
Boötes I	control	0.33	0.21	adiabatic
Fornax	control	0.14	0.16	adiabatic
Sculptor	control	—	0.12	adiabatic
Draco	control	—	0.07	adiabatic

Crater II is the prime carrier: extraordinarily diffuse (half-light radius ~ 1.1 kpc, the lowest internal density in the sample), eccentric ($e = 0.71$), with a *large* pericenter (24 kpc). **Antlia II** is the largest-radius dwarf known (~ 2.9 kpc), low density, eccentric ($e = 0.56$), pericenter 38 kpc. Both reach $y \sim 2.5\text{--}3.3$ at pericenter — squarely non-adiabatic. Both have large pericenters, which is a genuine design strength: they sit *away* from the tidal-stripping regime, so the prediction’s two real carriers are **tide-clean by construction** (Sec. 6). Everything denser — Fornax, Sculptor, Draco, the compact ultra-faints — stays adiabatic ($y < 0.3$) and acts as the control locus. The sample is thus sharply **bimodal**: a 2-object carrier set against a ~ 22 -object adiabatic background. This bimodality is the central fact governing what existing data can and cannot say.

5. The pre-registered pilot: a null at low power, reported straight

Before any fitting, I pre-registered the test (the registration, sign convention, primary statistic, and the prime confound are recorded in the data script; see Reproducibility). The hypotheses:

- **H1 (framework):** at fixed pericenter and mass, residual internal σ correlates **positively** with orbital eccentricity, predicted amplitude $+12\text{--}14\%$ (preferred; $+12\text{--}30\%$ envelope) radial-vs-circular at matched pericenter.
- **H0 (modified gravity / CDM):** **zero** partial correlation.

Data are a single homogeneous source — Pace, Erkal & Li 2022 (ApJ 940, 136; Gaia EDR3 proper motions) for *all* σ_{los} and *all* orbital parameters — minimizing cross-catalog systematics. $N = 24$ usable Milky-Way dwarfs (both σ and eccentricity measured). The pre-specified primary statistic is the partial Spearman correlation of σ_{los} with eccentricity, controlling for pericenter, a luminosity-based mass proxy, and half-light radius.

Result. The point estimate is negative and insignificant in every variant:

variant	ρ	p (two-sided)	reading
PRIMARY partial $\rho(\sigma, \text{ecc} \mid r_{\text{peri}}, L, r_{\text{half}})$	-0.196	0.395 (dof = 19)	null, wrong-signed
virial-deviation $\rho(\log \sigma, \text{ecc} \mid \sqrt{L/r_{\text{half}}}, r_{\text{peri}})$	-0.220	0.313	null
simple Spearman(σ, ecc)	-0.113	0.598	null
no-LMC eccentricities	-0.030	0.899	even flatter

The correlation is small, slightly **negative** (the *opposite* sign to the predicted positive effect), and statistically indistinguishable from zero (all p between 0.40 and 0.90). Adding the pre-registered tidal proxy $M_{\text{MW}}(<r_{\text{peri}})/r_{\text{peri}}^3$ as an extra control leaves it essentially unchanged: $\rho = -0.196$, $p = 0.408$. The result is **robust to the tidal-heating control** — but only because there is no signal for tides to displace.

I report this as what it is: a null with almost no power, not a strike against the theory. Three reasons, all physical rather than a matter of bad luck:

1. **The axis is wrong for existing data.** The framework’s signal lives on y , gated by diffuseness; only **two** of 24 dwarfs (Crater II, Antlia II) reach the carrier band. A population partial correlation on *raw eccentricity* is a weak surrogate — with 2 carriers buried in 22 adiabatic controls it cannot deliver the carrier-vs-control y -contrast the prediction actually makes.
2. **The σ errors are comparable to the predicted signal** (~10–40%, and *largest* on the carriers — Antlia II is 5.71 ± 1.08 km/s). The +12–14% effect is at or below this floor per object.
3. **An artifact pushes the carriers slightly cold.** The carriers sit modestly below a naive luminosity/ r_{half} virial baseline (Crater II $\sigma_{\text{dev}} = -0.36$, Antlia II -0.06). This is partly mechanical, not evidence against: y is itself built from σ (a low- σ diffuse dwarf is automatically high- y), so y and a σ -residual anti-correlate by construction. The clean, pre-registered eccentricity-axis test is the one to read — and it is null.

So: not suggestive (no surviving positive correlation), not tidal-confounded (robust to the control), simply **underpowered**. **Existing literature data does not yet show the effect, and existing data was never going to.** This pilot establishes only that fact; it does not, and cannot, move the theoretical status of the prediction. The sign remains a theorem.

6. The decisive test and its one real confound

The right test is **not** a raw-eccentricity population correlation. It is a **carrier-vs-control y -contrast** with the discriminating axis measured per object. Three ingredients make it decisive, all arriving on a near-term timeline:

1. **Gaia DR4 (December 2026)** sharpens per-dwarf proper motions, hence pericenter and the orbital frequency ω_{ext} , tightening y object-by-object — the axis is *measured*, not statistically inferred (the decisive improvement over the earlier, low-purity cluster-UDG version of the same physics, where infall phase had to be guessed at ~ 0.4 purity).
2. **A larger diffuse-carrier set** — Antlia II, Crater II, plus newly-characterized large-radius low-density satellites from DES/LSST follow-up — beats down per-object noise by turning a 2-point contrast into a population.
3. **Resolved σ profiles** (Keck/VLT/MUSE multi-object spectroscopy) rather than a single global σ .

The one real confound is tidal heating. A radial plunger with a small pericenter is also tidally stirred, which would *itself* inflate σ and could mimic the framework’s positive signal. The design separates the two:

- The prediction is the σ excess *at fixed pericenter* — tidal heating is controlled by regressing on pericenter (and $r_{\text{peri}}/r_{\text{tidal}}$ where available).
- Tides predict a **different radial σ profile** and a host-aligned stripping/elongation morphology, separable with resolved kinematics — whereas the bath-response effect is a global re-deepening into deep-MOND with no preferred tidal axis.
- The two named carriers are **tid-clean by construction**: Crater II and Antlia II have large pericenters (24, 38 kpc), sitting away from the stripping regime, yet are non-circular. They are non-adiabatic *without* being strongly tidally heated — exactly the clean corner the population test should target.

Run this way — carrier-vs-control y -contrast, Gaia DR4 orbits, resolved profiles, explicit tidal modeling — the test discriminates modified inertia from modified gravity at a sign level: a positive carrier-vs-control σ excess at matched pericenter is required by the framework and forbidden to all of metric/field MOND and CDM.

7. Scope and honest limitations

- **Pre-registered and falsifiable.** The hypotheses, sign convention, primary statistic, and prime confound were fixed before fitting. A null carrier-vs-control y -contrast at DR4 precision, or a *negative* one surviving the tidal control, would count against the framework’s modified-inertia content.
- **Modified-gravity-impossible.** The signal’s existence — not merely its size — is forbidden to every metric/field MOND and to CDM (Milgrom 2022). This is what makes it a discriminator, joining **Cassini** as a clean modified-inertia-vs-modified-gravity test.
- **The sign and the $y=1$ zero-crossing are theorems; the magnitude is now bath-constrained.** Plunge $\rightarrow$ hotter follows from θ decreasing through $\theta(1)=1$ alone. The de Sitter-Unruh bath now *selects* $\theta(0)=\sqrt{2}$ (the amplitude/excess-heat branch the framework already requires for the correct deep-MOND law) and *forces* a Lorentzian kernel shape from the de Sitter correlator, giving **+12–14%** (preferred) with a **+12–30%** envelope — replacing the earlier free-function “factor- ~ 2 .” The lone surviving modeling choice is the kernel’s corner *location*

(placed at ω_{internal} , quasi-statically licensed by Milgrom 1994, not derived); this is honestly *not* claimed as fully forced.

- **Near-term.** The decisive data (Gaia DR4) arrive December 2026; the carriers are named and the spectroscopy largely exists.
- **What this is not.** It is **not a TOE** — the framework does not derive the Standard Model and is a one-parameter effective theory at a frontier. It is **not a detection** — the present pilot is an explicit null at low power. It is **not a claim that existing dwarf data shows the effect** — they do not, and could not.

The honest standing: a sharp, MG-forbidden, pre-registered prediction with a measured discriminating axis and a named carrier set; a current null that reflects the data’s lack of power rather than the theory’s failure; and a decisive near-term test that the framework will pass or fail cleanly.

Reproducibility

All numbers in Sec. 5 reproduce from two scripts in `real_research/reviews/`:

- `dwarf_ecc_sigma_pilot_data.py` — the 24-dwarf data dictionary (every value carries a per-object source citation; σ_{los} and orbits from Pace, Erkal & Li 2022 / Gaia EDR3; Simon 2019 and McConnachie 2012 as cross-checks), the pre-registered hypotheses, sign convention, primary statistic, and the tidal confound control. Footing: $a_0 = 9.36 \times 10^{-11} \text{ m/s}^2$, framework interpolation only.
- `dwarf_ecc_sigma_pilot_analysis.py` — the per-dwarf $y = \omega_{\text{ext}}/\omega_{\text{in}}$ computation and the partial-correlation test.

Supporting derivations: `real_research/NATIVE_CONSEQUENCES_HORIZON_INERTIA_2026-06.md` (the prediction’s derivation from the bath premise), `real_research/DWARF_ECC_SIGMA_PILOT_2026-06.md` (the full pilot verdict), and `real_research/THETA_KERNEL_TOWARD_FORCED_2026-06.md` (the v_2 kernel derivation: $\theta(0)=\sqrt{2}$ selected from the framework’s excess-heat/amplitude coupling, the Lorentzian shape forced by the de Sitter Wightman function, with the corner location the lone surviving postulate).

References (key): Milgrom 1994 (modified-inertia must be time-nonlocal); Milgrom 2022 (the kernel $\theta(y)$, and the instantaneous external-field effect of modified gravity); Deser & Levin 1997 (de Sitter–Unruh temperature); Pace, Erkal & Li 2022, ApJ 940, 136 (σ and Gaia EDR3 orbits); Simon 2019, ARA&A 57, 375; McConnachie 2012, AJ 144, 4.

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